Rewarding the Buried Tonne: What Indonesia Can Learn from the EU's CCS-ETS Link

Bridging the policy gap to give Indonesian emitters a reason to store carbon at home

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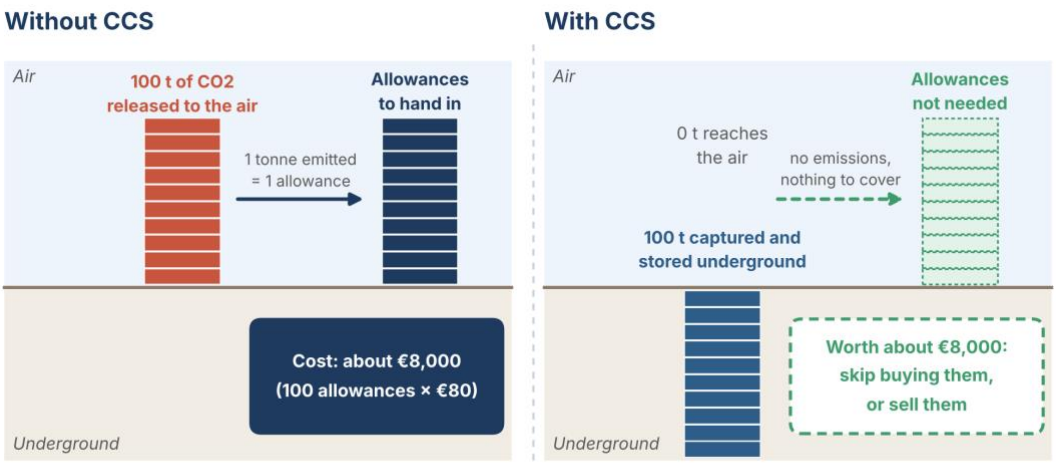
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KEY POINTS

- Indonesia has approximately **600 Gt** of potential CO2 storage, yet one of the clearest signs of demand so far is an MoU with Singapore. The open question is what gives Indonesian emitters a reason to capture and store CO2 at home.
- In the EU, CO2 captured and stored in a permitted site needs no allowances, so each stored tonne is worth about the allowance price (around **€84** in early 2026). This is what makes CCS appealing.
- The link works because the ETS Directive and the CCS Directive are written to fit together: the carbon market recognises only storage in sites permitted under the CCS rules.
- Indonesia's Presidential Regulations 14/2024 (CCS) and 110/2025 (carbon value) do not connect. No rule says a stored tonne reduces an installation's obligations, and the credit for a stored tonne has no stated owner.

Only a few countries are better placed to store carbon dioxide than Indonesia. Ministry of Energy and Mineral Resources calculations put its potential at about 600 gigatonnes, almost all of it in saline aquifers, which one official has said could hold the country's emissions for over three centuries [1]. Yet one of the clearest signals of demand so far comes from abroad: in June 2025, Indonesia and Singapore signed a memorandum of understanding on cross-border CCS [2]. At home, the open question is what would make an emitter choose to pay for carbon capture in the first place.

Storage space is only half of a market. The other half is a reason to use it. The European Union answered that question by tying its carbon market to its CCS rules, so that a tonne buried underground becomes a tonne an emitter no longer has to pay for. What follows traces how that link works, why it has turned CCS into a financial decision in Europe, and what Indonesia could take from it.



1 block = 10 tonnes of CO2, or 10 allowances. Illustration at €80 per allowance, before capture, transport and storage costs.

Figure 1. One cement plant emitting 100 tonnes of CO2, without and with CCS (illustrative).

Source: author's illustration, using an allowance price of about €80 in line with the early-2026 range in [3] and the EU treatment of stored CO2 described in [4] and [6]. Capture, transport and storage costs are not included.

1. Why a buried tonne is worth money

Start with the basic bargain of a carbon market. A government sets a limit on how much CO2 its big industries may emit in total, and issues permits, called allowances, up to that limit. Each allowance covers one tonne. An emitter that releases a hundred tonnes must hand in a hundred allowances at the end of the year, and because the total is capped and shrinking, allowances carry a price. In the EU, benchmark prices were around €84 in early 2026, after reaching nearly €94 in January [3].

As an example, for a cement factory, that price turns every tonne of CO2 into a bill, as the left side of Figure 1 shows. At roughly €80 each, a hundred tonnes cost about €8,000 in allowances (an illustration, before capture costs). Now suppose the plant captures that CO2 and stores it underground. The gas never reaches the air, so the site has no emissions to cover, and the allowances it would have handed in stay in its hands. It can skip buying them, or sell them to an emitter that needs them. These are two ways of cashing in the same allowance, not two separate payments. Either way, a captured and stored tonne is worth about the price of an allowance, so CCS makes financial sense whenever capture, transport and storage cost less than that.

2. How the EU wrote the link into law

That arithmetic works in Europe because two pieces of legislation were written to fit together. The ETS Directive, which runs the carbon market, treats capture, transport and storage of CO2 as part of the market itself [4]. The CCS Directive sets the rules for storage sites and issues their permits [5]. Between them, as the Commission's own analysis explains, an emitter does not have to hand in allowances for CO2 that is captured and stored in line with the EU's CCS rules [6].

The condition sits exactly where the two directives meet: the CO2 must go to a site holding a permit under the CCS Directive. The same analysis notes that only sites in the European Economic Area can obtain one, so storing the gas elsewhere does not earn the same treatment [6]. The permit is what allows the carbon market to treat a buried tonne as a tonne that was never emitted.

The bookkeeping follows the gas. Each installation along the chain, from the emitter that captures the CO2 to the pipeline or ship that moves it and the operator that stores it, adds the CO2 it receives to its own figures and subtracts what it passes on, so the numbers stay consistent from one end of the chain to the other [6]. Because every member state applies the same two directives, the arrangement works the same way whether the storage site lies next door or across a border.

The result is a pairing: a price that rewards a stored tonne, and a rule that lets the reward be claimed. Remove either and the case weakens. A price without the rule gives nothing back for storing, and a rule without a meaningful price saves too little to matter.

3. Where Indonesia stands

Indonesia has written rules for both halves, but in separate documents. Presidential Regulation 14/2024 governs CCS. It classes CCS as a climate mitigation action and requires the quantity stored to be measured (Articles 48 and 56), but when it comes to carbon value it points to "the regulation on carbon economic value" without naming one (Article 54) [7]. Presidential Regulation 110/2025 builds a carbon market with caps, quotas, offsets and credits, and leaves the sector minister to designate which installations are covered (Article 62), though none are named. It never mentions CCS [8].

What neither text contains is the rule at the centre of the EU model: nothing says that a tonne stored underground reduces what an installation has to cover. The nearest route runs through credits. An operator could earn a credit for a certified reduction and sell it to a covered installation (Articles 65 and 66 of Regulation 110/2025) [8], but the texts do not connect that route to storage, and they do not say who would earn the credit. The emitter and the storage operator are different parties (Article 47(6) of Regulation 14/2024), and the same regulation names the operator as the party responsible for the mitigation action (Article 48) [7]. That points toward the operator, though the text never says so. If the party paying for capture is not the party collecting the reward, the incentive may never reach the emitter that has to decide.

The same regulation shows where attention has gone so far. It reserves 70% of storage capacity for domestic CO2 and allows up to 30% for foreign CO2 (Article 35), and it makes cross-border storage depend on bilateral agreements (Articles 45 to 47) [7]. Space is set aside for home emitters, but the rule that would give them a reason to use it has not been written. And because Indonesia and its neighbours do not share a single rulebook the way EU members do, each cross-border deal has to be negotiated from scratch, which makes the domestic link the first one worth getting right.

4. What Indonesia could take from this

The EU example suggests four steps, offered here as proposals rather than findings:

- **Choose a route.** A stored tonne can either reduce what an installation must cover, as in the EU, or earn a credit that can be sold. Either could work, but the choice should be written down.
- **Say who benefits.** Where the emitter pays for capture and the storage operator receives the credit, the decision to invest sits with the party that gains least.
- **Give the rule a price to work with.** The link is worth only as much as the allowance behind it, so a trading system with no named sectors and a thin price leaves it little to build on.

Storage space can be surveyed and mapped. A reason to use it has to be written into law.

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